

# **LUNG NEOPLASMS AND PLUERAL LESIONS**

# OUTLINE

- ❖ Etiology and pathogenesis of lung cancer
- ❖ Classification of lung cancer
- ❖ Clinical course of lung cancer
- ❖ Neuroendocrine Proliferations and Tumors
- ❖ Miscellaneous Tumors
- ❖ Pleural Effusion
- ❖ Pleural Tumors

# Objectives

1. Recognize general feature of lung cancer
2. Describe etiology and pathogenesis of lung cancer
3. Describe the types and morphology of lung cancers
4. describe the paraneoplastic syndrome in lung carcinoma
5. Describe etiology and pathogenesis of plueral effusions

# Tumors

- ❖ A variety of benign and malignant tumors may arise in the lung, 90% to 95% are carcinomas, about 5% are carcinoid tumors, and 2% to 5% are mesenchymal and other miscellaneous neoplasms.

# Carcinomas

- ❖ Is the Most frequently diagnosed major cancer
- ❖ Most common cause of cancer mortality worldwide
- ❖ It is generally a disease of older adults, most often b/n ages 55 and 84 years

# Etiology and Pathogenesis.

- Most (but not all) lung cancers are associated with cigarette smoke.
- In addition, there are other genetic and environmental factors

## ☐ **Tobacco Smoking**

- 87% of lung carcinomas occur in active smokers or those who stopped recently.
- There is a nearly linear correlation between the frequency of lung cancer and pack-years of cigarette smoking.

# Etiology and Pathogenesis.

- The increased risk is 60 times greater in habitual heavy smokers(two packs a day for 20 years) than in nonsmokers
- lung cancer develops in only 10% to 15% of smokers
- Women have a higher susceptibility to tobacco carcinogens than men do.
- Although cessation of smoking decreases the risk for lung cancer over time, it may never return to baseline levels
- Chewing tobacco is not a safe substitute for smoking cigarettes or cigars >> **spare the lung but cause oral cancers and can lead to nicotine addiction.**
- electronic cigarette ??

# Etiology and Pathogenesis.

- *Secondhand smoke*, or environmental tobacco smoke, contains numerous human carcinogens for which there is no safe level of exposure
- **What of heavy smokers who never develop cancer?**
  - the mutagenic effect of carcinogens in smoke is modified by genetic variants
  - mutagen sensitivity genotype
- basal cell hyperplasia and squamous metaplasia progress to squamous dysplasia and carcinoma in situ then to invasive carcinoma.

# Etiology and Pathogenesis.

## ❑ Industrial Hazards.

- Certain industrial exposures, such as arsenic, chromium, uranium, nickel, vinyl chloride ,etc increase the risk of developing lung cancer
- High-dose ionizing *radiation* is carcinogenic
- Asbestos exposure also increases the risk for lung cancer development.
  - The latent period is 10 to 30 years
  - Lung cancer is the most frequent malignancy in individuals exposed to asbestos, particularly when coupled with smoking

## ❑ Air Pollution

- It is uncertain whether it increases the risk of lung cancer by itself
- Chronic exposure > lung irritation, inflammation, and repair
- radon gas

## ❑ ***Acquired Mutations***

- smoking-related carcinomas of the lung arise by a stepwise accumulation of oncogenic “driver” mutations
- field effect
- ❖ The major histologic subtypes of lung cancer each have distinctive molecular features, as follows:
  - **Adenocarcinoma** -is associated with tobacco smoking, but less so than other histologic subtypes;
    - as a result, it is the most common subtype in never-smokers .
    - About one-third of adenocarcinomas have oncogenic gain-of-function mutations involving components of growth factor receptor signaling pathways (EGFR, ALK...)
    - Other tumors have gain-of-function mutations in serine/threonine kinases (*BRAF*, 2% of tumors, and *PI3K*, 2% of tumors) or in the *KRAS* gene (roughly 30% of tumors)

- **Squamous cell carcinoma**

- is highly associated with exposure to tobacco smoke and
- harbors diverse genetic aberrations, many of which are chromosome deletions involving tumor suppressor loci. These losses, especially those involving 3p, 9p (*CDKN2A* gene), and 17p (*TP53* gene)

- **Small cell carcinoma**

- is virtually always smoking related
- has the highest mutational burden among lung cancers.
- There is almost universal inactivation of both *TP53* and *RB*,
- unusual transformations of non–small cell carcinoma to small cell carcinoma are often associated with acquisition of *RB* loss-of-function mutations

- ***Lung Cancer in Never-Smokers***

- WHO estimates that 25% of lung cancer worldwide occurs in never-smokers.
- occur more commonly in women
- most are adenocarcinomas, often with targetable mutations/co-mutations
- are more likely to have *EGFR* mutations and **almost never have *KRAS* mutations**

## ❑ ***Precursor (Preinvasive) Lesions.***

- (1) Atypical adenomatous hyperplasia,
  - (2) adenocarcinoma in situ,
  - (3) squamous dysplasia and carcinoma in situ, and
  - (4) Diffuse idiopathic pulmonary neuroendocrine cell hyperplasia.
- ❖ the term *precursor* does not imply that progression to cancer is inevitable.
  - ❖ Currently it is not possible to distinguish between precursor lesions that progress and those that remain localized or regress

# Classification

- ❖ Tumor classification is important for consistency in patient treatment and because it provides a basis for epidemiologic and biologic studies.

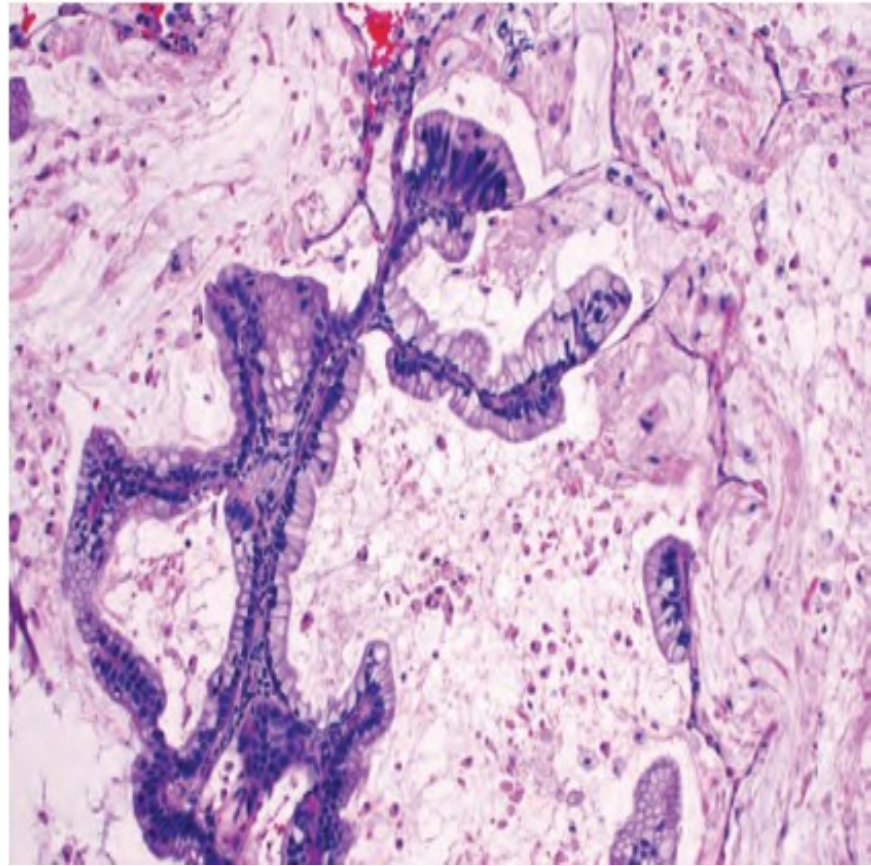
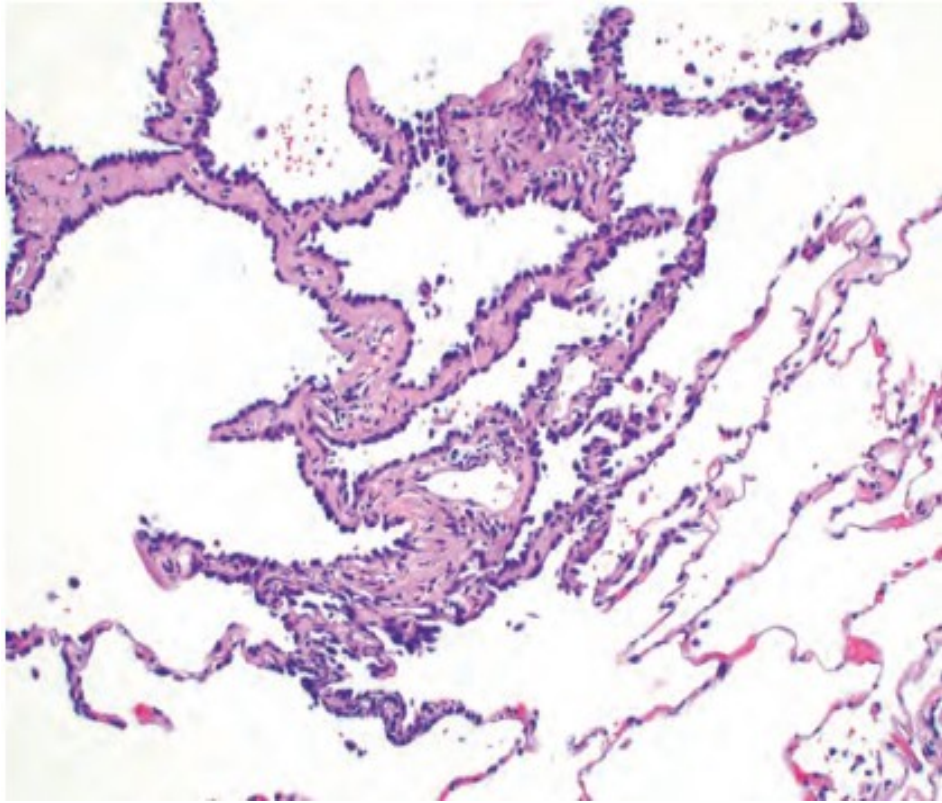
# Classification.

- The relative proportions of the major categories are:
  - Adenocarcinoma (50%)
  - Squamous cell carcinoma (20%)
  - Small cell carcinoma (15%)
  - Large cell carcinoma (2%)
  - Other (13%)

**Table 15.9 Histologic Classification of Malignant Epithelial Lung Tumors**

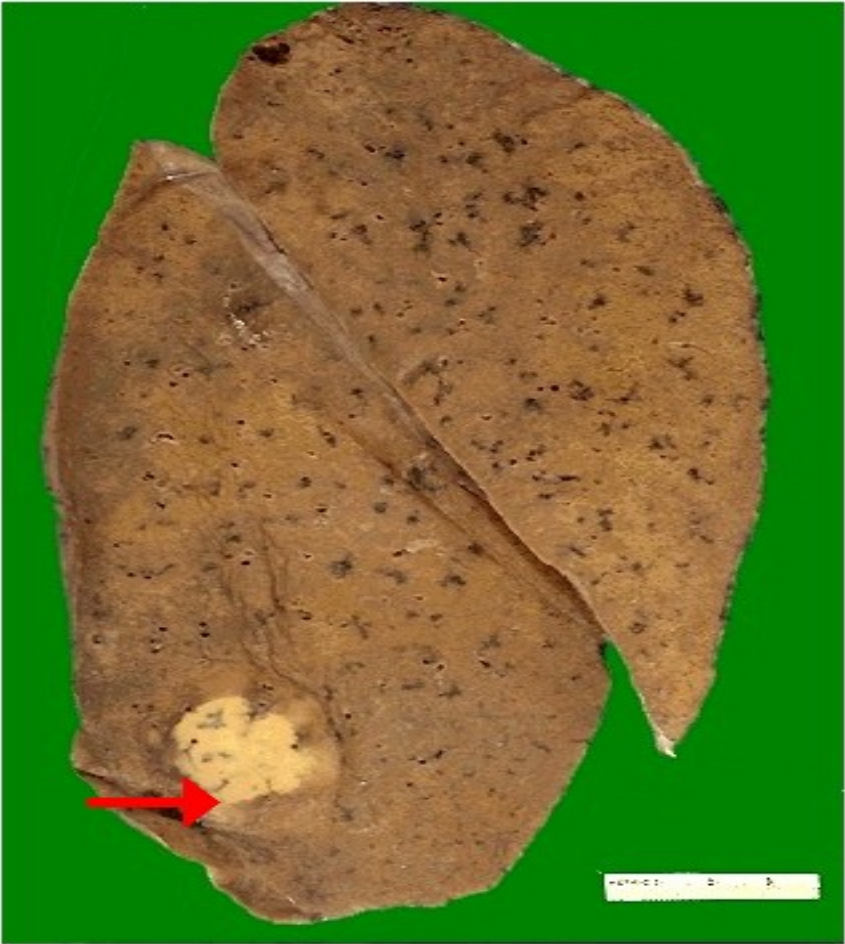
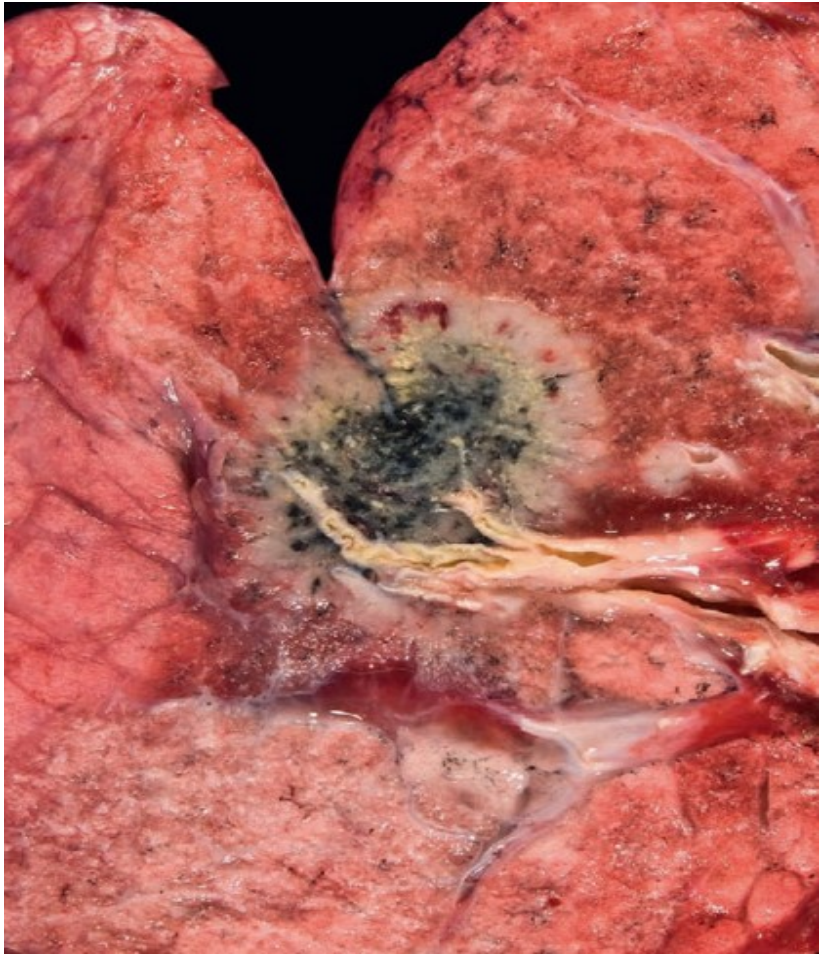
| <b>Tumor Classification</b>                                                          |
|--------------------------------------------------------------------------------------|
| <b><i>Adenocarcinoma</i></b>                                                         |
| Lepidic, acinar, micropapillary, papillary, solid (according to predominant pattern) |
| Invasive mucinous adenocarcinoma                                                     |
| Minimally invasive adenocarcinoma (nonmucinous, mucinous)                            |
| <b><i>Squamous cell carcinoma</i></b>                                                |
| Keratinizing, nonkeratinizing, basaloid                                              |
| <b><i>Neuroendocrine tumors</i></b>                                                  |
| Small cell carcinoma                                                                 |
| Combined small cell carcinoma                                                        |
| Large cell neuroendocrine carcinoma                                                  |
| Combined large-cell neuroendocrine carcinoma                                         |
| Carcinoid tumor                                                                      |
| Typical, atypical                                                                    |
| <b><i>Other uncommon types</i></b>                                                   |
| Large cell carcinoma                                                                 |
| Adenosquamous carcinoma                                                              |
| Sarcomatoid carcinoma                                                                |
| Pleomorphic, spindle cell, giant cell carcinoma, carcinosarcoma, pulmonary blastoma  |
| Others such as lymphoepithelioma-like carcinoma and NUT carcinoma                    |
| Salivary gland–type tumors                                                           |

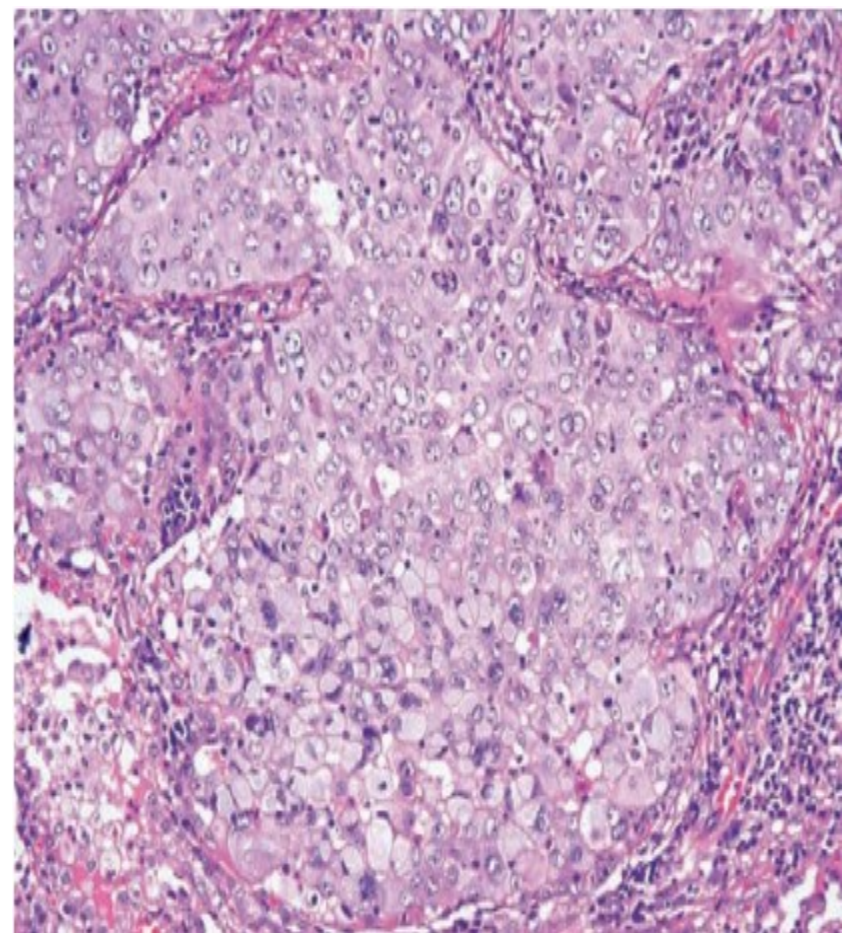
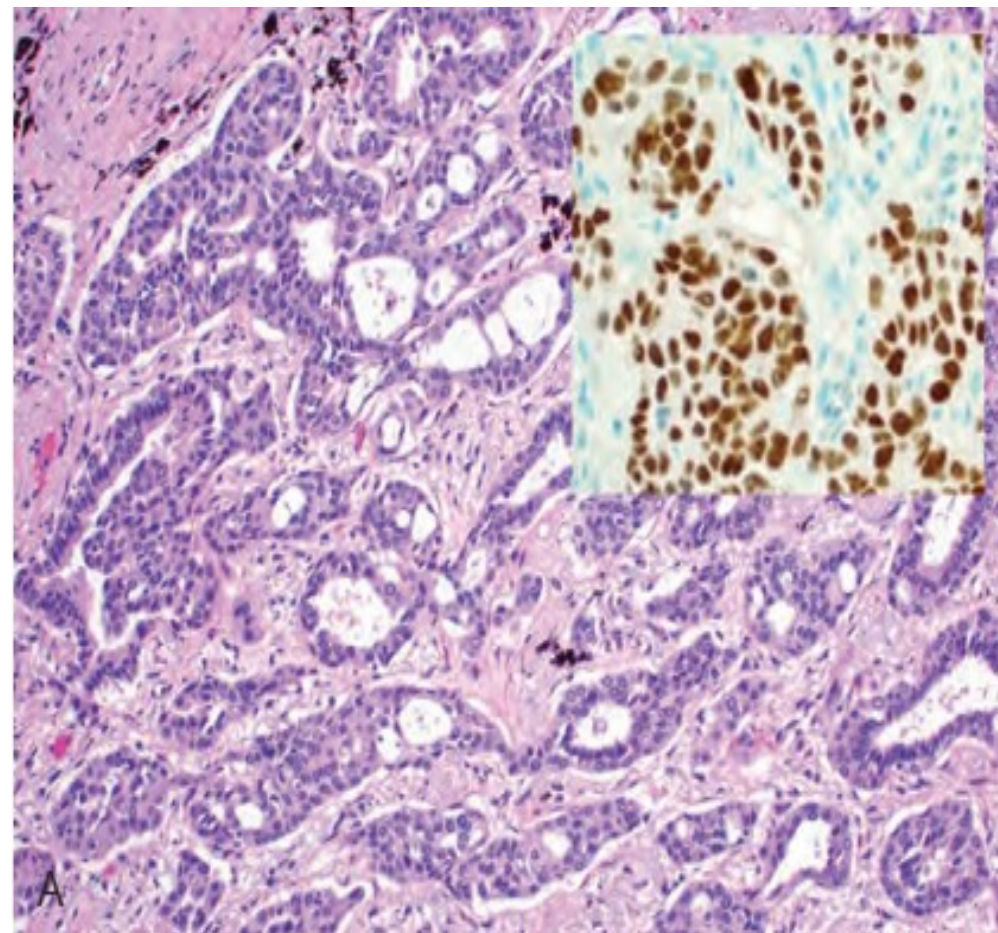
- **Atypical adenomatous hyperplasia**
  - Is a small precursor lesion ( $\leq 5$  mm) xed by dysplastic pneumocytes lining alveolar walls that are mildly fibrotic.
  - It can be single or multiple and
  - can be in the lung adjacent to invasive tumor or away from it.
- **Adenocarcinoma in situ**
  - is a lesion that is less than 3 cm in size and is composed entirely of dysplastic cells growing along pre-existing alveolar septa.
  - The cells have more dysplasia than atypical adenomatous hyperplasia and may or may not have intracellular mucin



# Adenocarcinoma

- is an invasive malignant epithelial tumor with glandular differentiation, mucin production, or pneumocyte marker expression
- Five histologic patterns (acinar, papillary, micropapillary, lepidic, solid); mucinous and non mucinous subtypes
- Positive for TTF1
- Upper lobe > lower lobe
- Peripheral > central
- tend to be smaller
- Tumors ( $\leq 3$  cm) with a small invasive component ( $\leq 5$  mm) associated with scarring and a peripheral lepidic growth pattern are called **microinvasive adenocarcinoma**

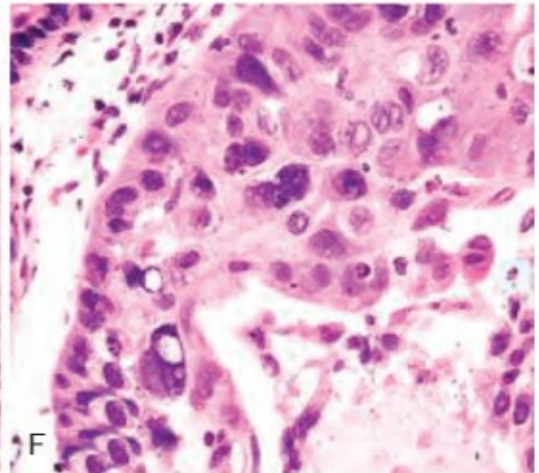
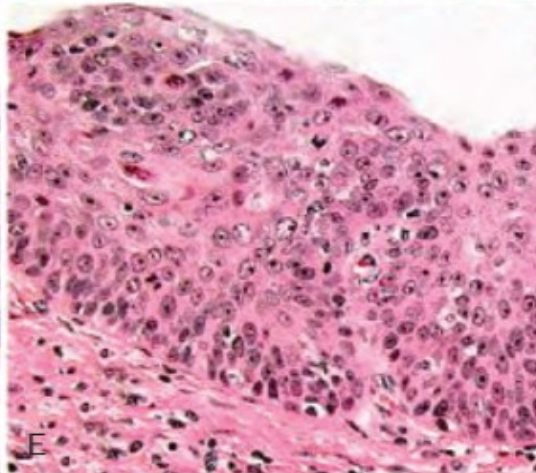
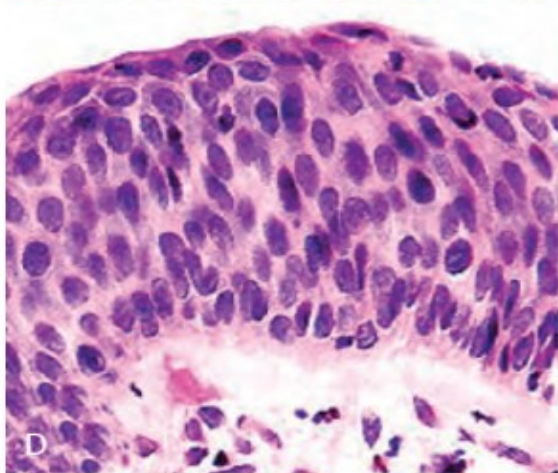
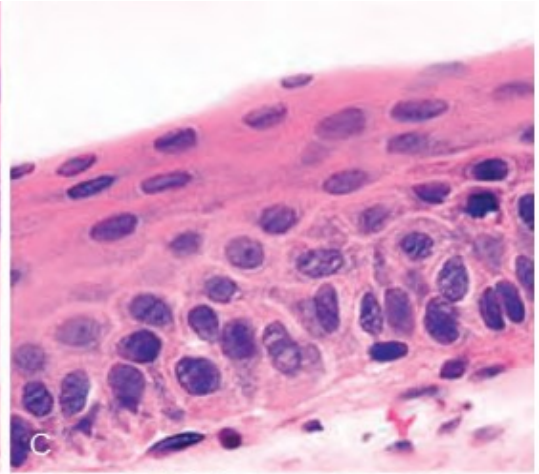
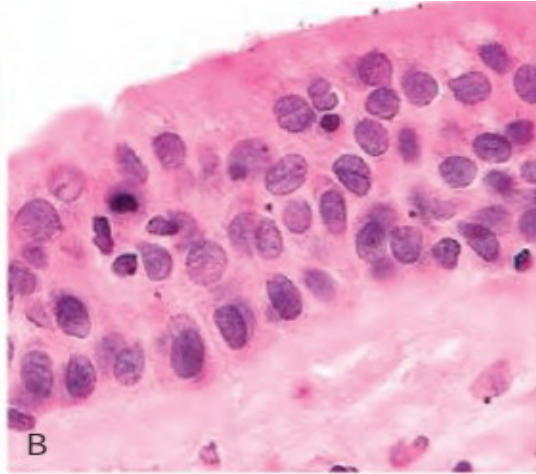
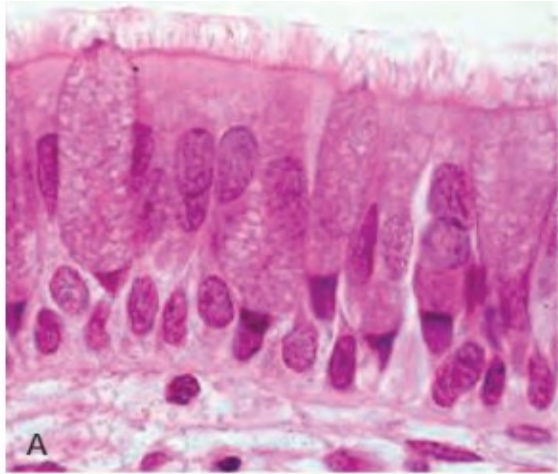


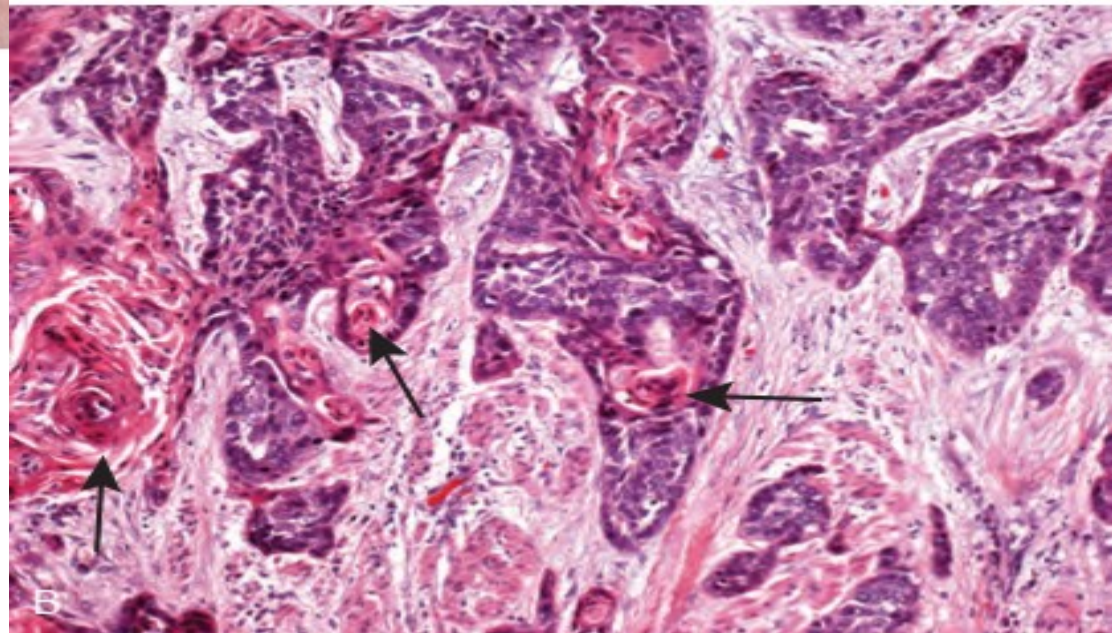
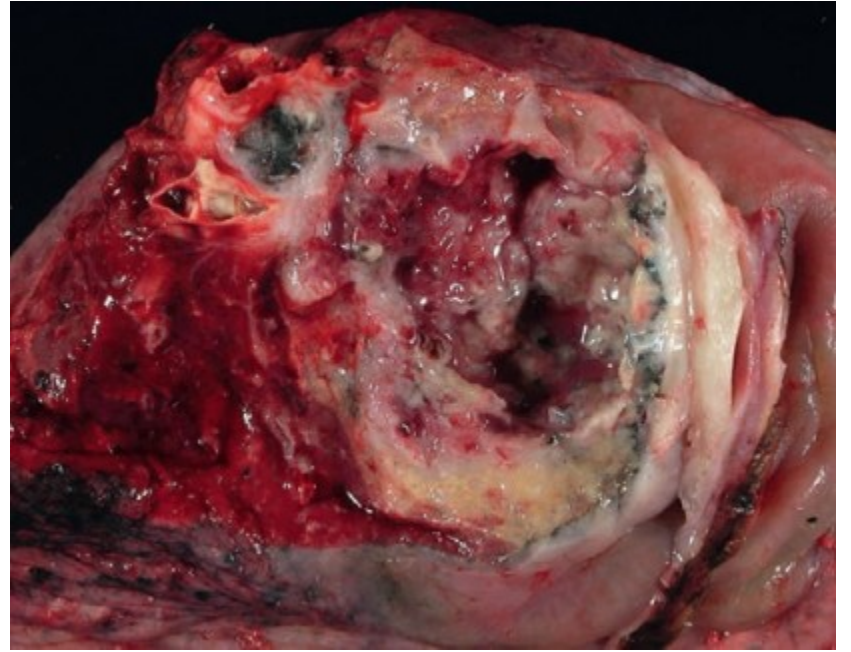
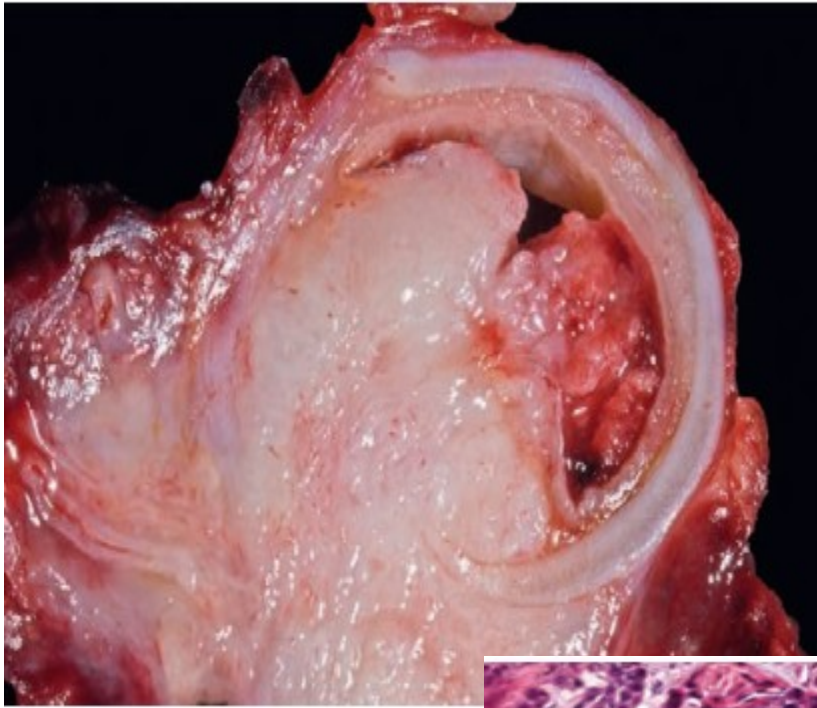


# ***Squamous Cell Carcinoma***

- is more common in men
- is strongly associated with smoking.
- It may grow exophytically into the bronchial lumen, producing an intraluminal mass.
- With further enlargement the bronchus becomes obstructed, leading to distal atelectasis and infection.
- The tumor may also penetrate the wall of the bronchus and infiltrate along the peribronchial tissue into the adjacent carina or mediastinum.

- In other instances, the tumor grows along a broad front to produce a cauliflower-like intraparenchymal mass that compresses the surrounding lung.
- Especially when the tumors are bulky, focal areas of hemorrhage or necrosis may appear to produce red or yellow-white mottling and softening. Sometimes these necrotic foci cavitate
- Histologically, squamous cell carcinoma is characterized by the presence of keratinization and/or intercellular bridges
- In the past, most SCCs arose centrally from the segmental or subsegmental bronchi, but the incidence of SCC of the peripheral lung is increasing

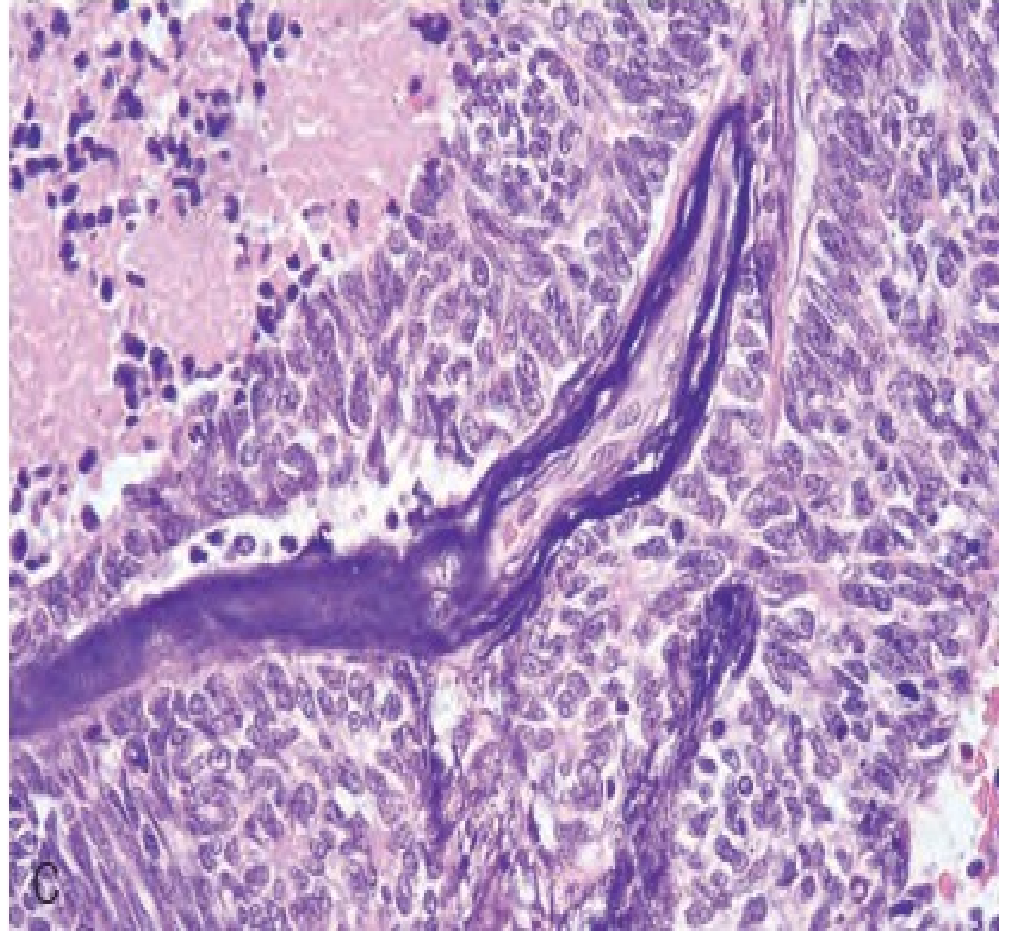




# ***Small Cell Carcinoma***

- is a highly malignant tumor with a strong relationship to cigarette smoking;
- only about 1% occurs in nonsmokers
- Typically central / bronchial / hilar; rarely a peripheral nodule
- Submucosal growth
- It may arise in major bronchi or in the periphery of the lung.
- There is no known pre-invasive phase.
- They are the most aggressive of lung tumors, metastasizing widely and virtually always proving to be fatal.
- is comprised of relatively small cells with scant cytoplasm, ill-defined cell borders, finely granular nuclear chromatin (salt and pepper pattern), and absent or inconspicuous nucleoli .

- The cells are round, oval, or spindle shaped,
- nuclear molding is prominent.
- There is no absolute size for the tumor cells, but in general they are **smaller than three times the diameter of a small resting lymphocyte** (a size of about 25  $\mu\text{m}$ ).
- The mitotic count is high.
- The cells grow in clusters that exhibit neither glandular nor squamous organization.
- Necrosis is common and often extensive.
- Azzopardi effect is frequently present.
- Combined small cell carcinoma

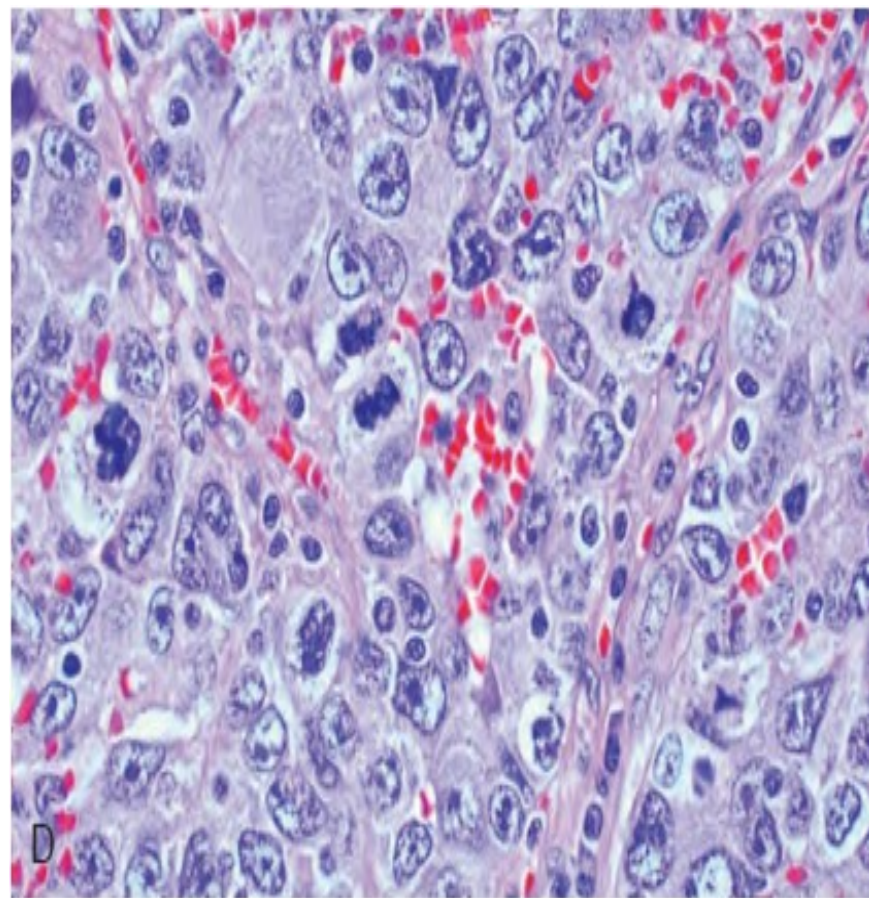
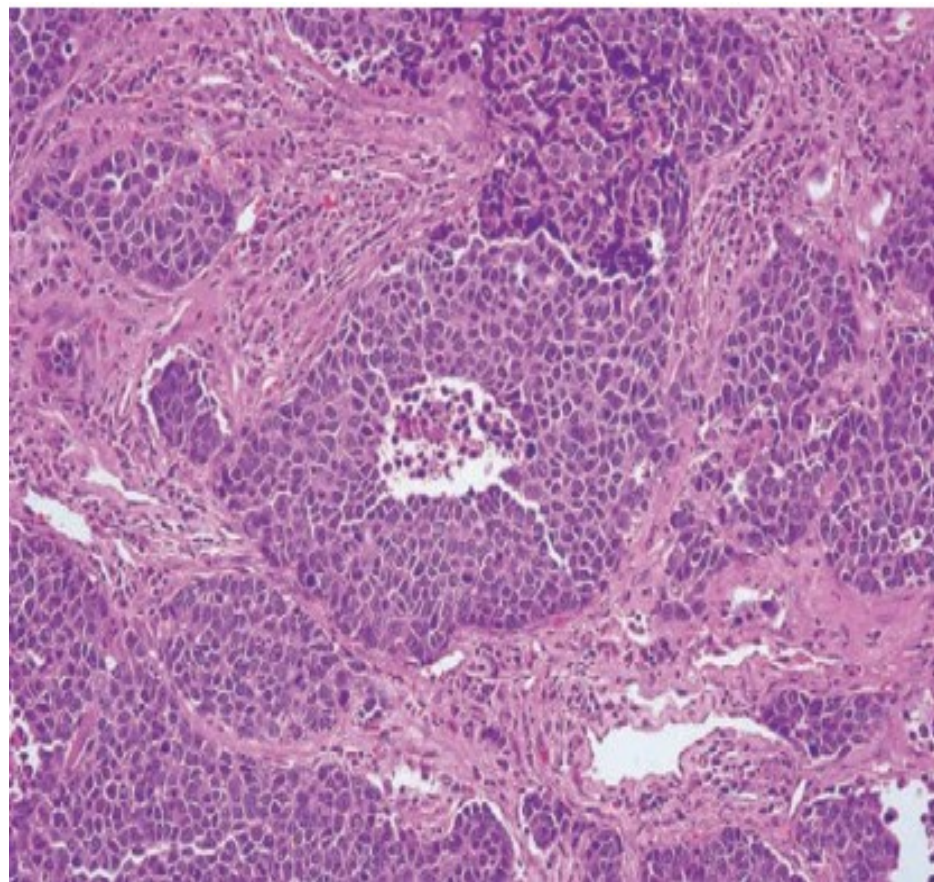


# large cell neuroendocrine carcinoma

- High grade non small cell carcinoma with neuroendocrine morphology and immunohistochemical markers, characterized by  $> 10$  mitoses /  $2\text{mm}^2$  and extensive necrosis
- Large cells ( $\sim 3\times$  size of small cell carcinoma) with abundant cytoplasm, variably coarse chromatin, nuclear pleomorphism, prominent nucleoli
- Prognosis is poor, similar to that of small cell carcinoma

# ***Large Cell Carcinoma***

- is an undifferentiated malignant epithelial tumor that lacks the cytologic features of other forms of lung cancer.
- The cells typically have large nuclei, prominent nucleoli, and a moderate amount of cytoplasm.
- It is a diagnosis of exclusion since it expresses none of the markers associated with adenocarcinoma (TTF-1, napsin A) or squamous cell carcinoma (p40, p63).



# Combined Carcinoma

- Approximately 4% to 5% of all lung carcinomas have a combined histology, including two or more of the aforementioned types

# Secondary Pathology

- **focal emphysema**
- **Atelectasis**
- **severe suppurative or ulcerative bronchitis or**
- **bronchiectasis**
- **Pulmonary abscesses**
- **superior vena cava syndrome**
- **pericarditis or pleuritis**
- **Horner syndrome** (enophthalmos, ptosis, miosis, and anhidrosis)

# Staging

- A uniform TNM system for staging cancer according to its anatomic extent at the time of diagnosis is extremely useful, chiefly for comparing treatment results from different centers .
- Stage I to IV.

# Clinical Course

- ❖ Lung cancer is one of the most insidious and aggressive neoplasms in the realm of oncology.
- ❖ Over 50s whose symptoms are of several months duration.
- ❖ *The major presenting complaints are cough (75%), weight loss (40%), chest pain (40%), and dyspnea (20%).*

**Table 15.11 Local Effects of Lung Tumor Spread**

| <b>Clinical Feature</b>            | <b>Pathologic Basis</b>                                                |
|------------------------------------|------------------------------------------------------------------------|
| Cough (50%–75%)                    | Involvement of central airways                                         |
| Hemoptysis (25%–50%)               | Hemorrhage from tumor in airway                                        |
| Chest pain (20%)                   | Extension of tumor into mediastinum, pleura, or chest wall             |
| Pneumonia, abscess, lobar collapse | Airway obstruction by tumor                                            |
| Lipoid pneumonia                   | Tumor obstruction; accumulation of cellular lipid in foamy macrophages |
| Pleural effusion                   | Tumor spread into pleura                                               |
| Hoarseness                         | Recurrent laryngeal nerve invasion                                     |
| Dysphagia                          | Esophageal invasion                                                    |
| Diaphragm paralysis                | Phrenic nerve invasion                                                 |
| Rib destruction                    | Chest wall invasion                                                    |
| SVC syndrome                       | SVC compression by tumor                                               |
| Horner syndrome                    | Sympathetic ganglia invasion                                           |
| Pericarditis, tamponade            | Pericardial involvement                                                |

- ❖ The best “treatment” for lung cancer is smoking prevention,
- ❖ overall 5-year survival rate is only 18.7%
- ❖ In general, adenocarcinoma and squamous cell carcinoma tend to remain localized longer and have a slightly better prognosis than small cell carcinoma, which is usually advanced by the time it is discovered
- ❖ Small cell carcinoma is quite sensitive to radiation therapy and chemotherapy, and approximately 10% of patients with limited disease survive for 5 years and may be cured

# Paraneoplastic Syndromes

- ❖ Lung carcinoma can be associated with several paraneoplastic syndromes, some of which may antedate the development of a detectable pulmonary lesion.
- ❖ The hormones or hormone-like factors elaborated include:
  - ❖ *Antidiuretic hormone* (ADH), inducing hyponatremia due to inappropriate ADH secretion
  - ❖ *Adrenocorticotrophic hormone* (ACTH), producing Cushing syndrome

# Paraneoplastic Syndromes.

- ❖ *Parathormone, parathyroid hormone-related peptide, prostaglandin E, and some cytokines*, all implicated in the hypercalcemia often seen with lung cancer
- ❖ *Calcitonin*, causing hypocalcemia
- ❖ *Gonadotropins*, causing gynecomastia
- ❖ *Serotonin and bradykinin*, associated with the carcinoid syndrome
- ❖ Tumors that produce ACTH and ADH are predominantly small cell carcinomas,
- ❖ tumors that produce hypercalcemia are mostly squamous cell carcinomas

# Neuroendocrine Proliferations and Tumors

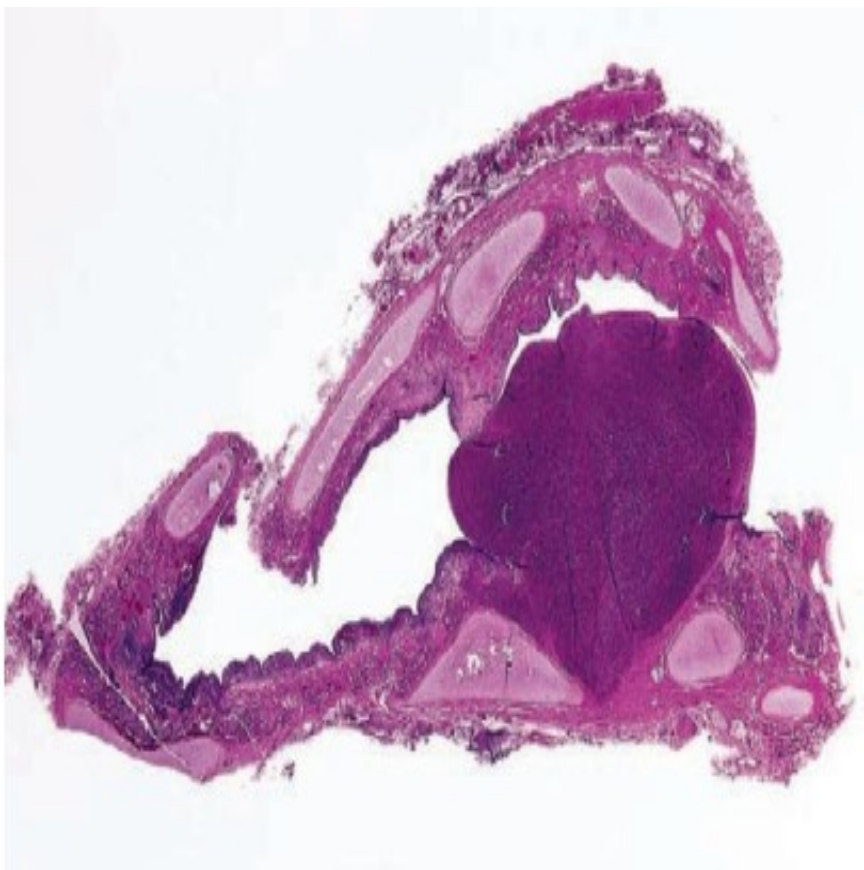
- Virtually all pulmonary neuroendocrine cell hyperplasias are secondary to airway fibrosis and/or inflammation.
- The exception is a rare disorder called *diffuse idiopathic pulmonary neuroendocrine cell hyperplasia*
- Neoplasms of neuroendocrine cells in the lung include benign *tumorlets*; *carcinoids*; and small cell carcinoma and large cell neuroendocrine carcinoma of the lung.
- Carcinoid tumors are classified separately, since they differ significantly from carcinomas with evidence of neuroendocrine differentiation in terms of incidence and clinical, epidemiologic, histologic, and molecular characteristics.
- For example, in contrast to small cell and large cell neuroendocrine carcinomas, carcinoids may occur in patients with multiple endocrine neoplasia type 1.

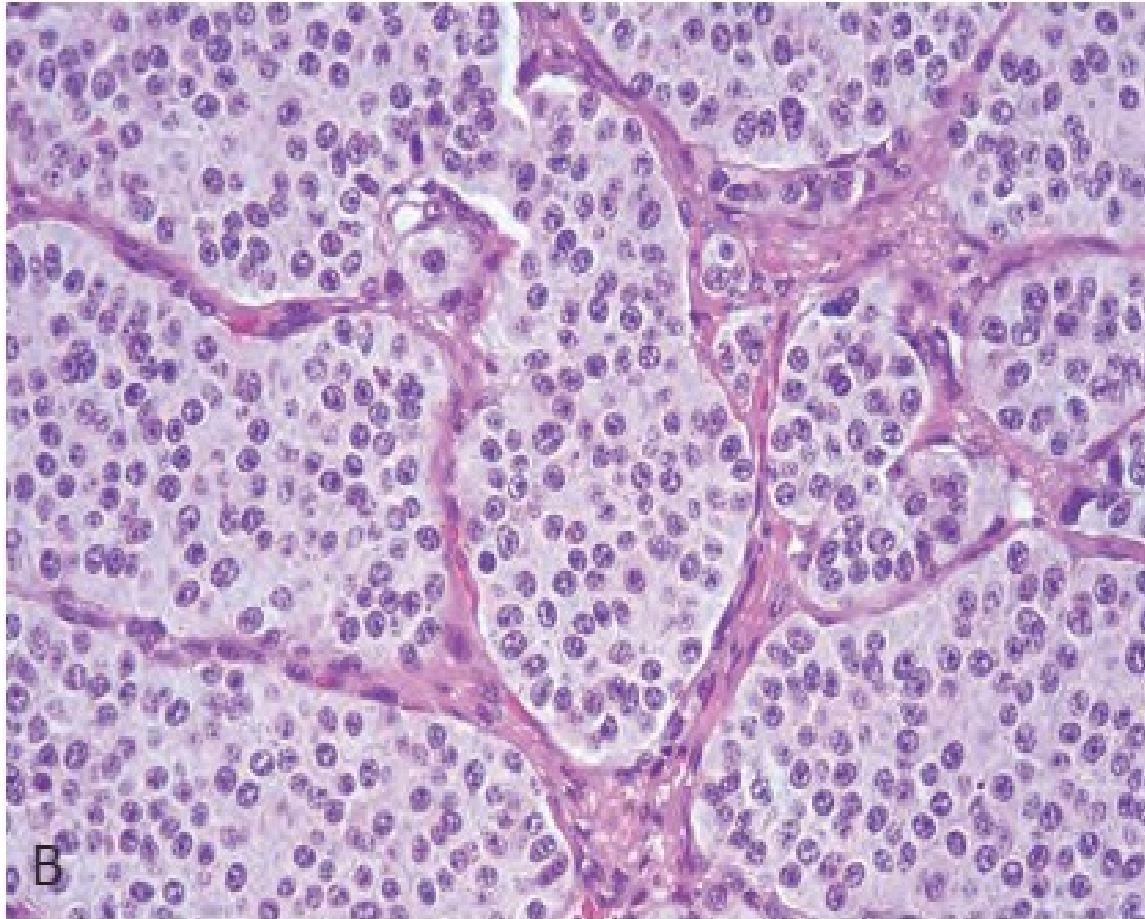
# Carcinoid Tumors

- represent 1% to 5% of all lung tumors.
- Most patients with these tumors are younger than 60 years of age,
- the incidence is equal for both sexes.
- App 20% to 40% of patients are nonsmokers.
- are low-grade malignant epithelial neoplasms that are subclassified into *typical* and *atypical carcinoids*
- may arise centrally or may be peripheral

- On gross examination, the central tumors grow as finger-like or spherical polypoid masses that commonly project into the lumen of the bronchus
- are usually covered by an intact mucosa
- They rarely exceed 3 to 4 cm in diameter.
- Most are confined to the mainstem bronchi.
- Others, however, penetrate the bronchial wall to fan out in the peribronchial tissue, producing the so-called **collar-button lesion**.
- Peripheral tumors are solid and nodular.

- Histologically, the tumor is composed of organoid, trabecular, palisading, ribbon, or rosette-like arrangements of cells separated by a delicate fibrovascular stroma.
- the individual cells are quite regular and have uniform round nuclei and a moderate amount of eosinophilic cytoplasm
- **Typical carcinoids**
  - is the most common of the neuroendocrine lung tumors
  - are well circumscribed with limited invasion of adjacent lung tissues
  - have <2 mitoses per 10 HPFs and lack necrosis,
- **atypical carcinoids**
  - Have 2-10 mitoses per 10 HPFs and/or foci of necrosis.
  - show increased pleomorphism,
  - have more prominent nucleoli, and
  - are more likely to grow in a disorganized fashion and invade lymphatics





# Metastatic Tumors

- **The lung is the most common site of metastatic neoplasms.**
- Both carcinomas and sarcomas arising anywhere in the body may spread to the lungs via the blood or lymphatics or by direct continuity
- In the usual case, multiple discrete nodules (cannonball lesions) are scattered throughout all lobes, particularly in the lung periphery.
- Other times spread takes the form of a solitary nodule, endobronchial or pleural involvement, pneumonic consolidation, and/or some combination thereof.
- Foci of lepidic growth similar to adenocarcinoma in situ are seen occasionally with metastatic carcinomas and may be associated with any of the listed patterns



# PLEURAL EFFUSION

- ❑ is a common manifestation of both primary and secondary pleural diseases, which may be inflammatory or noninflammatory.
- ❑ Normally, no more than 15 mL of serous, relatively acellular, clear fluid lubricates the pleural surface.

# PLEURAL EFFUSION

❑ **Accumulation of pleural fluid occurs in the following settings:**

- Increased hydrostatic pressure
- Increased vascular permeability
- Decreased osmotic pressure
- Increased intrapleural negative pressure
- Decreased lymphatic drainage

# Inflammatory Pleural Effusions

- Serous, serofibrinous, and fibrinous pleuritis all have an inflammatory basis, differing only in the intensity and duration of the process
- The common causes of pleuritis are inflammatory diseases within the lungs, such as tuberculosis, pneumonia, lung infarcts, lung abscess, and bronchiectasis.

# Inflammatory Pleural Effusions

- Rheumatoid arthritis, disseminated lupus erythematosus, uremia, diffuse systemic infections, other systemic disorders, and metastatic involvement of the pleura can also cause serous or serofibrinous pleuritis
- Radiotherapy for tumors in the lung or mediastinum often causes a serofibrinous pleuritis

# Inflammatory Pleural Effusions

- ❑ A purulent pleural exudate (*empyema*) usually results from bacterial or mycotic seeding of the pleural space.
- ❑ Most commonly, this seeding occurs by contiguous spread of organisms from intrapulmonary infection
- ❑ occasionally, it occurs through lymphatic or hematogenous dissemination from a more distant source.

# Inflammatory Pleural Effusions

- ❑ ***hemorrhagic pleuritis*** manifested by sanguineous inflammatory exudates is infrequent and is found in hemorrhagic diatheses, rickettsial diseases, and neoplastic involvement of the pleural cavity.
  - The sanguineous exudate must be differentiated from hemothorax.
  - When hemorrhagic pleuritis is encountered, careful search should be made for the presence of exfoliated tumor cells
- ❑ **hemothorax**

# Noninflammatory Pleural Effusions

- Noninflammatory collections of serous fluid within the pleural cavities are called *hydrothorax*.
- The fluid is clear and straw colored.
- Hydrothorax may be unilateral or bilateral, depending on the underlying cause.

## ❑ **Common causes**

- cardiac failure
- renal failure and
- cirrhosis

# ***Chylothorax***

- ❑ *Chylothorax* is an accumulation of milky fluid, usually of lymphatic origin, in the pleural cavity.
- ❑ Chyle is milky white because it contains finely emulsified fats.
- ❑ Is most often caused by thoracic duct trauma or by obstruction of a major lymphatic duct, usually by a malignancy

# PNEUMOTHORAX

- refers to air or gas in the pleural cavities
- is most commonly associated with emphysema, asthma, and tuberculosis
- It may be spontaneous, traumatic, or therapeutic
- Spontaneous pneumothorax may complicate any form of pulmonary disease that causes emphysematous changes.
- An abscess cavity that communicates either directly with the pleural space or with the lung interstitial tissue may also lead to the escape of air

- **spontaneous idiopathic pneumothorax**
- **tension pneumothorax**
  - the pleural defect acts as a flap valve and permits the entrance of air during inspiration but fails to permit its escape during expiration

# PLEURAL TUMORS

- ❑ The pleura may be involved by primary or secondary tumors.
- ❑ Secondary metastatic involvement is far more common than are primary tumors.
- ❑ The most frequent metastatic malignancies arise from primary neoplasms of the lung and breast.

# Solitary Fibrous Tumor

- ❑ Soft-tissue tumor with a propensity to occur in the pleura and, less commonly, in the lung, as well as other sites.
- ❑ The tumor is often attached to the pleural surface by a pedicle.
- ❑ Small to large
- ❑ it tends to remain confined to the surface of the lung
- ❑ Grossly, it consists of dense fibrous tissue with occasional cysts filled with viscid fluid

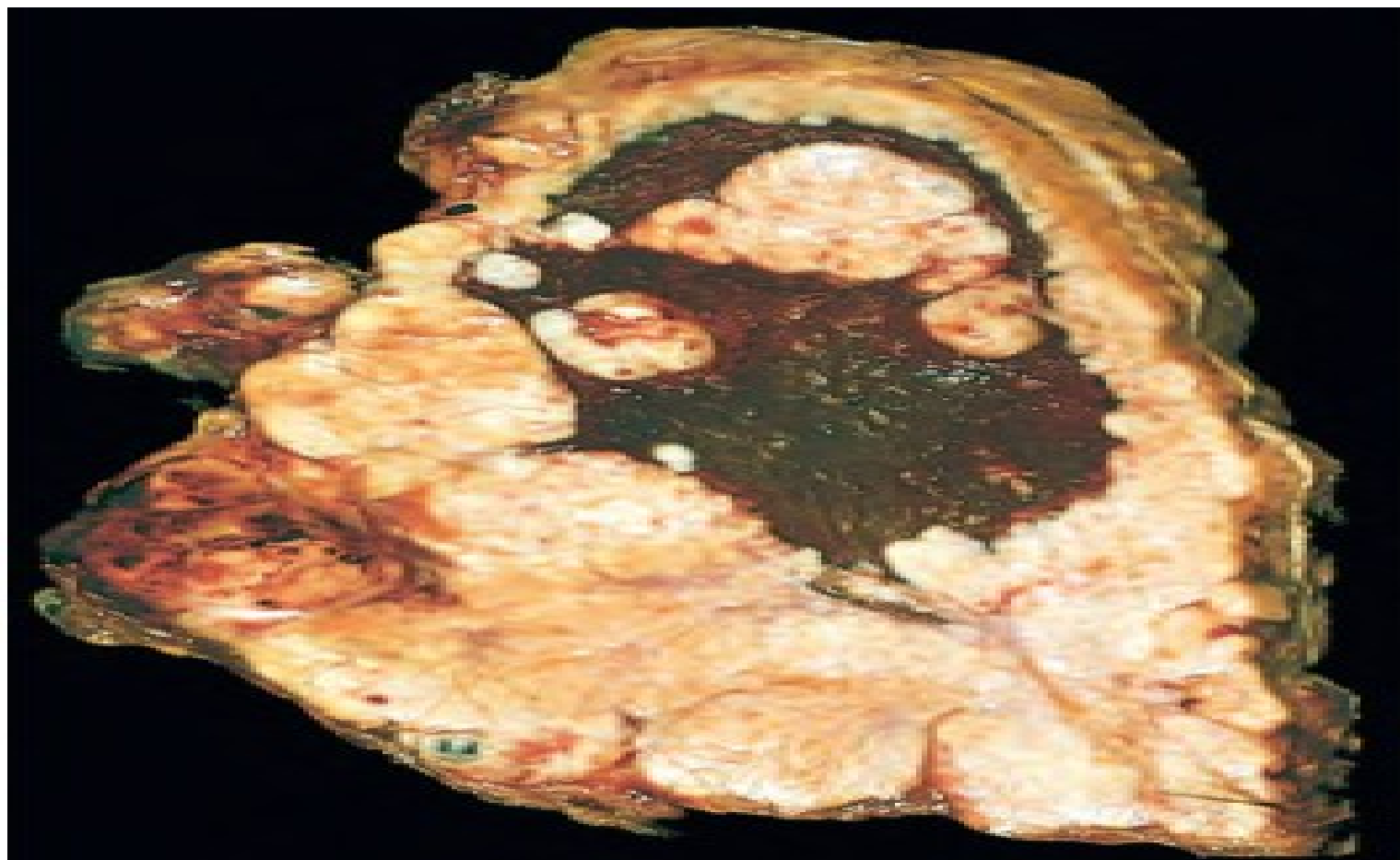
# Solitary Fibrous Tumor

- ❑ Microscopically, the tumor shows whorls of reticulin and collagen fibers among which are interspersed spindle cells resembling fibroblasts.
- ❑ Rarely, this tumor may be malignant, with pleomorphism, mitotic activity, necrosis, and large size (>10 cm).

# Malignant Mesothelioma

- ❑ Malignant mesotheliomas in the thorax arise from either the visceral or the parietal pleura.
- ❑ Though uncommon, they have assumed great importance in the past few years because of their increased incidence among people with heavy exposure to asbestos
- ❑ The lifetime risk of developing mesothelioma in heavily exposed individuals is as high as 7% to 10%.
- ❑ There is a long latent period of 25 to 45 years for the development of asbestos-related mesothelioma

- ❑ the most common cytogenetic abnormalities is homozygous deletion of chromosome 9p(*CDKN2A*)-in about 80% of mesotheliomas
- ❑ Malignant mesothelioma is a diffuse lesion that spreads widely in the pleural space and is usually associated with extensive pleural effusion and direct invasion of thoracic structures.
- ❑ The affected lung becomes ensheathed by a thick layer of soft, gelatinous, grayish pink tumor tissue



# Morphology

- ❑ Microscopically, malignant mesothelioma may be epithelioid (60% to 80%), sarcomatoid (10% to 12%), or biphasic (10% to 15%)
- ❑ This is in keeping with the fact that mesothelial cells have the potential to develop as epithelium-like cells or mesenchymal stromal cells.

# Clinical Course.

- ❑ The presenting complaints are chest pain, dyspnea, and, as noted, recurrent pleural effusions.
- ❑ Concurrent pulmonary asbestosis (fibrosis) is present in only 20% of individuals with pleural mesothelioma.
- ❑ The lung is invaded directly, and there is often metastatic spread to the hilar lymph nodes and, eventually, to the liver and other distant organs
- ❑ Mesotheliomas also arise in the peritoneum, pericardium, tunica vaginalis, and genital tract

**END**